

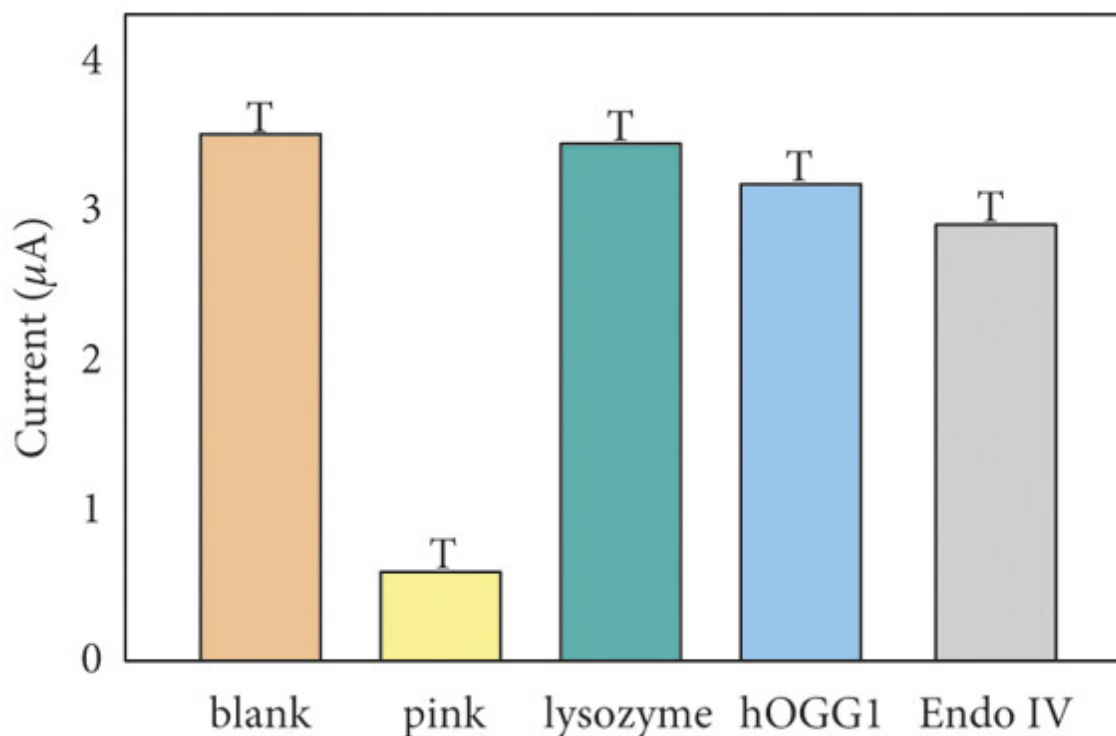
PSYC 6136: Assignment 1

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DDAR Exercise 1.1

a) *“Examine a number of recent issues of a scientific or statistical journal in which you have some interest. Find one or more examples of a graph or table that is a particularly bad use of display material to summarize and communicate research findings. Write a few sentences indicating how or why the display fails and how it could be improved.”*



Retrieved from: <https://doi.org/10.1155/2022/3802603>

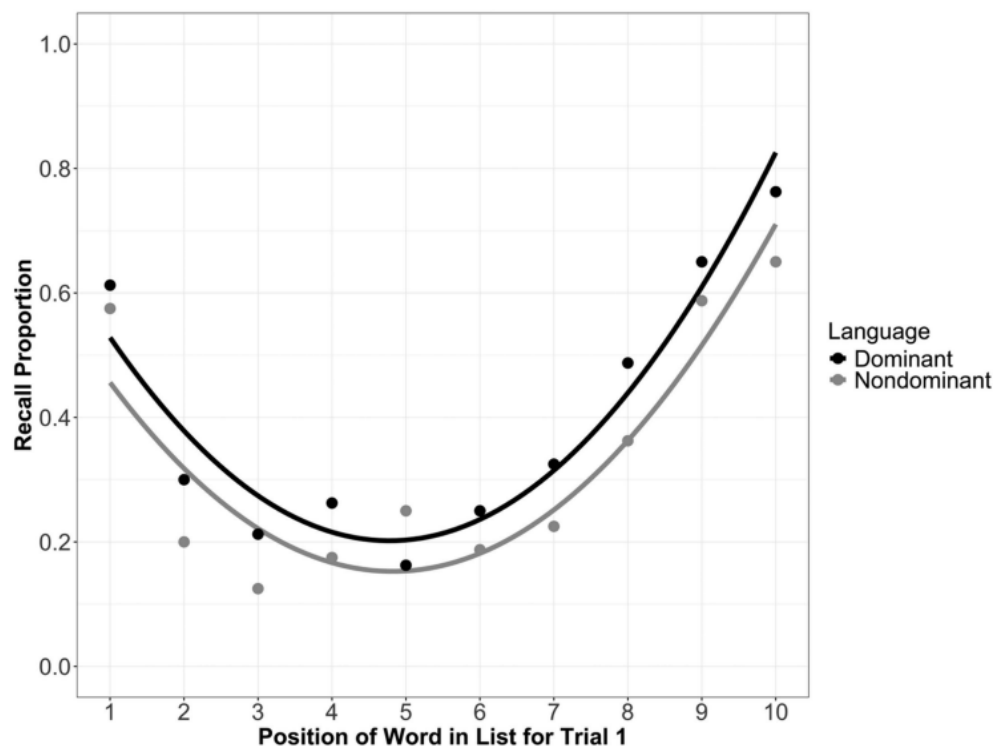
This graph is from a (now retracted) paper that was published in *Advances in Materials Science and Engineering* in 2022. It attempts to depict the selectivity of a biosensor with several enzymes interfering. From the text, it is not clear what this graph attempts to show and why “pink” deviates from the rest of the enzymes. Additionally, “pink” and “blank” are not enzymes. An x -axis label would help to clear the confusion.

regarding “pink” and “blank.” Furthermore, the “error bars” are actually the letter “T” superimposed on the graph. Authors should have used a statistical measure of uncertainty (e.g., standard error) to construct proper error bars. Last, the graph would benefit from a title and a better explanation in the figure caption (as is, the caption simply reads “Selectivity of the analytical method.”).

DDAR Exercise 1.2

“As in the previous exercise, examine the recent literature in recent issues of some journal of interest to you. Find one or more examples of a graph or table that you feel does a good job of summarizing and communicating research findings.”

Figure 4
Recall Accuracy by Serial Position for 40 Spanish–English Bilinguals



Retrieved from: <https://doi.org/10.1037/neu0001058>

a) *“Write a few sentences describing why you chose these displays.”*

This graph published in *Neuropsychology* in 2025 shows how a participant’s ability to recall a word changes depending on both the position of the word in a list and whether the language was his/her dominant or non-dominant. From the graph, primacy and recency effects on recall are indicated through the higher recall proportion for the first few and last few words in the list, with the minimum recall proportion being located near the middle of the list. We may also see that recall proportion is higher if the words were in the participant’s dominant language. A proper title and axes labels were included.

b) *“Now take the role of a tough journal reviewer. Are there any features of the display that could be modified to make them more effective?”*

This graph can be modified to directly label the dominant and non-dominant curves with text, as opposed to the inclusion of a legend. It might also benefit from a figure caption to assist the reader in quickly understanding the information conveyed.

DDAR Exercise 2.2

“For each of the following data sets in the `vcdExtra` package, identify which are response variable(s) and which are explanatory. For factor variables, which are unordered (nominal) and which should be treated as ordered? Write a sentence or two describing substantive questions of interest for analysis of the data. (Hint: use `data(foo, package="vcdExtra")` to load, and `str(foo)`, `help(foo)` to examine data set `foo`.)”

a) *“Abortion opinion data”*

```
data(Abortion, package="vcdExtra")
?vcdExtra::Abortion
```

1. `Support_Abortion` is a nominal response variable
2. `Status` is an ordinal explanatory variable
3. `Sex` is a nominal explanatory variable

Do sex and/or status meaningfully predict whether one supports abortion?

b) *“Caesarian births data”*

```
data(Caesar, package="vcdExtra")
?vcdExtra::Caesar
```

1. `Infection` is a response variable. If a Type 1 infection is better/worse than a Type 2 infection, `Infection` may be treated as ordered upon severity. Else, `Infection` is nominal.
2. `Risk` is a nominal explanatory variable
3. `Antibiotics` is a nominal explanatory variable
4. `Planned` is a nominal explanatory variable

Does the presence of risk factors, being given antibiotics, and/or having planned the C section meaningfully predict infection absence/type?

c) *“Dayton Survey data”*

```
data("DaytonSurvey", package="vcdExtra")
?vcdExtra::DaytonSurvey
```

1. `cigarette` is a nominal response variable
2. `alcohol` is a nominal response variable
3. `marijuana` is a nominal response variable
4. `sex` is a nominal explanatory variable
5. `race` is a nominal explanatory variable

Do sex and/or race meaningfully predict whether high school seniors have ever used alcohol, cigarettes, and/or marijuana?

d) “*Minnesota high school graduates data*”

```
data(Hoyt, package="vcdExtra")
?vcdExtra::Hoyt
```

1. `Status` is a nominal response variable
2. `Rank` is an ordinal explanatory variable
3. `Occupation` is an ordinal explanatory variable
4. `Sex` is a nominal explanatory variable

Do sex, one’s rank in his/her graduating class, one’s father’s occupational status, and/or one’s sex meaningfully predict one’s post high-school status?

DDAR Exercise 2.4

“The data set DanishWelfare in vcd gives a 4-way, $3 \times 4 \times 3 \times 5$ table as a data frame in frequency form, containing the variable Freq and four factors, Alcohol, Income, Status and Urban. The variable Alcohol can be considered as the response variable, and the others as possible predictors.”

a) “*Find the total number of cases represented in this table*”

```
data("DanishWelfare", package = "vcd")
(sum(DanishWelfare$Freq))
```

```
## [1] 5144
```

b) “*In this form, the variables Alcohol and Income should arguably be considered ordered factors. Change them to make them ordered.*”

```
df <- DanishWelfare

df$Alcohol <- ordered(df$Alcohol,
                      levels = c("<1", "1-2", ">2"))
df$Income <- ordered(df$Income,
                     levels = c("0-50", "50-100", "100-150", ">150"))
```

c) “Convert this data frame to table form, `DanishWelfare.tab`, a 4-way array containing the frequencies with appropriate variable names and level names.”

```
DanishWelfare.tab <- with(df, xtabs(Freq ~ Alcohol + Income + Status + Urban))
structable(Alcohol ~ Income + Status + Urban, data = DanishWelfare.tab)
```

			Alcohol	<1	1-2	>2
##						
##	Income	Status	Urban			
##	0-50	Widow	Copenhagen	1	3	2
##			SubCopenhagen	4	0	0
##			LargeCity	1	1	2
##			City	8	4	1
##			Country	6	2	0
##		Married	Copenhagen	14	15	1
##			SubCopenhagen	8	7	2
##			LargeCity	41	15	2
##			City	100	25	7
##			Country	175	48	7
##		Unmarried	Copenhagen	6	2	3
##			SubCopenhagen	1	3	0
##			LargeCity	2	9	1
##			City	6	9	5
##			Country	9	7	1
##	50-100	Widow	Copenhagen	8	1	3
##			SubCopenhagen	2	1	0
##			LargeCity	7	3	2
##			City	14	8	1
##			Country	5	4	3
##		Married	Copenhagen	42	39	14
##			SubCopenhagen	51	59	21
##			LargeCity	62	68	14
##			City	234	172	38
##			Country	255	143	35
##		Unmarried	Copenhagen	7	12	2
##			SubCopenhagen	5	3	0
##			LargeCity	9	11	3
##			City	20	20	12
##			Country	27	23	13
##	100-150	Widow	Copenhagen	2	5	2
##			SubCopenhagen	3	4	1
##			LargeCity	1	1	1
##			City	5	9	1
##			Country	2	4	0
##		Married	Copenhagen	21	32	20
##			SubCopenhagen	30	68	31
##			LargeCity	23	43	10
##			City	87	128	36
##			Country	77	86	21
##		Unmarried	Copenhagen	3	6	0
##			SubCopenhagen	2	10	2
##			LargeCity	1	5	3
##			City	12	21	9
##			Country	4	15	7

```
## >150   Widow   Copenhagen      42  26  21
##                               SubCopenhagen    29  34  13
##                               LargeCity         17  14   5
##                               City              95  48  20
##                               Country           46  24   8
##      Married Copenhagen      24  43  23
##                               SubCopenhagen    30  76  47
##                               LargeCity         50  70  21
##                               City             167 198  53
##                               Country          232 136  36
##      Unmarried Copenhagen     33  36  38
##                               SubCopenhagen    24  23  20
##                               LargeCity         15  48  13
##                               City              64  89  39
##                               Country           68  64  26
```

d) “The variable Urban has 5 categories. Find the total frequencies in each of these. How would you collapse the table to have only two categories, City, Non-city?”

```
# Total frequencies in each Urban category
margin.table(DanishWelfare.tab, "Urban")
```

```
## Urban
##      Copenhagen SubCopenhagen   LargeCity      City      Country
##           552           614           594      1765      1619
```

```
# Collapse Urban to two categories: City, Non-city
twoCat <- collapse.table(DanishWelfare.tab,
                          Urban = c("City", "Non-city", "City", "City", "Non-city"))
```

e) “Use structable() or ftable() to produce a pleasing flattened display of the frequencies in the 4-way table. Choose the variables used as row and column variables to make it easier to compare levels of Alcohol across the other factors.”

```
structable(Alcohol ~ Income + Status + Urban, data = twoCat)
```

```
##                               Alcohol  <1 1-2 >2
## Income Status Urban
## 0-50   Widow   City      10   8   5
##        Non-city      10   2   0
##        Married  City     155  55  10
##        Non-city     183  55   9
##        Unmarried City     14  20   9
##        Non-city     10  10   1
## 50-100 Widow   City     29  12   6
##        Non-city      7   5   3
```

##	Married	City	338	279	66
##		Non-city	306	202	56
##	Unmarried	City	36	43	17
##		Non-city	32	26	13
##	100-150	Widow	8	15	4
##		Non-city	5	8	1
##	Married	City	131	203	66
##		Non-city	107	154	52
##	Unmarried	City	16	32	12
##		Non-city	6	25	9
##	>150	Widow	154	88	46
##		Non-city	75	58	21
##	Married	City	241	311	97
##		Non-city	262	212	83
##	Unmarried	City	112	173	90
##		Non-city	92	87	46